

pharmaceuticals. The company in question had to use live and genetically engineered cells while having to track about 200 variables so that they can ensure the purity of the manufacturing process. The resultant products were vaccines and blood components.



Mass scale production is affected by too many parameters to be tracked traditionally

Now, the fascinating thing was that two separate batches of the same substance manufactured using the same process resulted in a yield variation from a whopping 50 to 100 percent! Such variations could result in attracting regulatory attention.

To tackle the problem the project team divided the manufacturing process into different activity clusters. They assessed the process interdependencies using big data analytics and were able to identify nine parameters which had a direct impact on the vaccine yield. Using the analysis the company successfully increased the production of vaccine by 50 percent. Something that gave the company annual savings between \$5 million and \$10 million.

2. Custom design of products

Consumerism is at an age when everything from advertising to the product design itself is getting more and more customised, catering to the individual tastes keeping in tune with the ethos of 'It's my life' that modern brands seem to be promoting. And companies are increasingly turning to big data to help customize product designs for their consumer.